LLDD Database - Target Schema and Data Dictionary

SBP Mall - ระบบประกันรายได้ | Low Level Design Document

แนวทางการใช้เอกสาร

หัวข้อ	คำอธิบาย
วัตถุประสงค์	ใช้เป็นรายละเอียดระดับพัฒนาสำหรับออกแบบ ลงมือพัฒนา ตรวจทาน และทดสอบขอบเขตที่ระบุในฉบับนี้
ลำดับการอ่าน	เริ่มจาก Overview และ Scope จากนั้นตรวจ Field/Validation, Implementation, Contract, Processing Flow, Acceptance Criteria และ Test Checklist ตามลำดับ
ผลลัพธ์ที่คาดหวัง	ผู้อ่านสามารถระบุ input, ขั้นตอนประมวลผล, output, เงื่อนไขผิดพลาด และหลักฐานการทดสอบได้จากเนื้อหาในฉบับเดียว

คำว่า Input, Progress และ Output ในเอกสารนี้หมายถึงข้อมูลตั้งต้น ลำดับการทำงาน
และผลลัพธ์ที่ตรวจสอบได้ของขอบเขตที่กำลังพัฒนา

1. Purpose

เอกสารนี้เป็น LLDD Database ระดับรวมของ target schema ระบบ SBPGI/SBP Mall ใช้เป็น reference สำหรับ BE API, Batch Job, migration, indexing, transaction และ data dictionary

2. Architecture Context

- ระบบใหม่รวม EAI และ K2 เข้าเป็น SBPGI ใช้ฐานข้อมูลเดียวกัน
- ไม่มีไฟล์ BPM06001O/BPM06002O/BPM06003O ภายในเพื่อส่งเข้า K2; ใช้ FK จาก compensation_documents ไป impact_process แทน
- Workflow engine ภายในใช้ workflow_instances และ workflow_tasks แทน K2 engine ภายนอก
- ตัดขั้นบัญชี 04/05 (SDD v7.5 — รวมเข้าการออกแบบแล้ว ไฟล์ต้นฉบับถูกลบจาก repo 2026-08-06); workflow ใช้ section 06/08/01/02/03; SDD ที่ยึดเป็นหลักคือ SDD GI 24/02/2026
- มาตรฐานชื่อ table/column เป็น English lower_snake_case
- ตาราง job_configs และ job_run_histories เป็น schema reference สำหรับ BE/dev; ไม่ใช่ scope ให้ FE Batch Monitor ทำ tab Database ที่ใช้

2.1 Input / Progress / Output Contract

Stage	Contract for implementation
Input	Target table catalog, data zones, primary keys, foreign-key relationships, migration assumptions, index needs, and transaction boundaries.
Progress	Use the data spine impact_process_id -> doc_no -> instance_id/task_id to implement APIs/jobs, then validate referential integrity, idempotency keys, and audit writes.
Output	Data dictionary and implementation reference for schema creation, migration, indexing, transaction handling, and test data preparation.

3. Data Zones and Spine

Zone	Scope	Core tables	Owner usage
A	FGI/FCS Impact Pipeline and external interfaces	fgi_impact_processes, fgi_impact_stores, sales, interface_transactions	Batch Jobs 1-7, IAS/ALLMAP/QSSI/STA tracking
B	K2 Document and internal workflow	compensation_documents, document_* tables, workflow_instances, workflow_tasks	Document APIs, workflow actions, FE detail/list/report
C	Shared master/config/RBAC/audit	stores, roles, menus, configs, jobs, email templates, audit	Lookup, admin, job monitor, notification

Order	Key	Meaning	Used by
1	impact_process_id	หนึ่งร้านถูกกระทบ + หนึ่งงวด	FGI/FCS pipeline, Job 8/8b
2	doc_no	เอกสาร YYYY/xxxxx ปี พ.ศ.	Document APIs, reports, attachments
3	instance_id	workflow instance ต่อเอกสาร	Workflow engine
4	task_id	งานต่อ section/assignee	Inbox/current task guard
5	employee_id / role_code	identity/RBAC	menu, audit, assignment

4. Data Dictionary

Zone	Table	PK	FK / relationship	Role
A	fgi_impact_stores	id	impact_process_id, impacted_store_code	impact pair; sales request and allocation data
A	fgi_impact_processes	id	impacted_store_code	impact process hub and canonical workflow_generation_status
A	fgi_impact_sales_summaries	id	impact_process_id	sales summary/growth rate
A	sales_transactions	id	sales_summary_id	daily sales 4 windows x 15 days
A	fgi_impact_competitors	id	impact_process_id	ALLMAP competitors
A	fcs_qssi_scores	id	store_id + category_code + period	QSSI scores
A	interface_transactions	id	impact_process_id/sales_summary_id/doc_no	interface tracking replacement
B	compensation_documents	doc_no	impact_process_id, status_code, current_section_code	document header/core
B	document_new_stores	id	doc_no, new_store_code	new stores, compensate percent and amount
B	document_competitors	id	doc_no, competitor_code	document competitors
B	document_external_factors	id	doc_no, factor_code	document external factors
B	consideration_logs	id	doc_no	approval/action history (decision code, result category, attachments)
B	document_attachments	attach_id	doc_no	attachment metadata; file storage uses existing SBP S3 service
B	compensation_histories	id	store_code, ref_doc_no	compensation history/accounting export
B	document_cost_details	id	doc_no, new_store_code	monthly cost detail per new store (ImpactCostDetail)
B	document_running_numbers	year	-	atomic YYYY/xxxxx running number
C	impacted_stores	store_code	store.store_code (SBP)	SP impacted store subset
C	decisions	decision_code	-	decision master (button/flow/result names)
C	external_factors	factor_code	-	external factor master
C	competitors	competitor_code	-	competitor master
C	audit_logs	id	table_name + ref_key	master/config/email audit
C	status_email_rules	status_code	workflow state (SBP)	notification recipients
C	job_configs	job_no	-	schema reference for batch schedule/config; not FE Database tab scope
C	job_run_histories	run_id	job_no	job execution history

Zone	Table	PK	FK / relationship	Role
-	USE EXISTING SBP TABLES	-	-	workflow_transaction/approver/history/state/route/status (@srm/glb-workflow) · store/mas_store/sevenshop · mas_zone · common_code · business_user · email_template + email_sent · mas_param — decided 2026-08-05/2026-08-06: do NOT recreate these in SBPGI

4.1 Canonical Column Contract

Table	Canonical columns used by DDL and SQL	Rejected legacy vocabulary
menu_permissions	role_code, menu_code, can_access	can_view/can_create/can_update/can_delete
user_accounts	employee_id, role_code, section_code, username, is_active	password_hash/account_status
workflow_instances	instance_id, doc_no, instance_status, started_at, started_by	status or auto-generated instance id
system_configs	config_key, category, config_value, value_type, unit, description, is_editable	secret_flag or inline secrets
sales_transactions	txn_date, window_no, sales_amount, sales_diff, is_outlier	sale_date/window_code/net_sales
consideration_logs	result, result_category, detail, consider_by, action_datetime	result_code/comment/considered_by/considered_at
interface_transactions	id, acked_at	tracking_id/receive_date (API aliases only)
fgi_impact_processes	workflow_generation_status	duplicate workflow flag on fgi_impact_stores

5. Executable DDL — 34 Tables

ส่วนนี้เป็น PostgreSQL DDL ครบทุกตารางใน Data Dictionary เรียงตาม dependency พร้อม PK, typed FK, unique/check constraint และ index ที่จำเป็น สามารถใช้เป็น migration baseline ได้โดยไม่ต้องเดา column เพิ่มเติม

5.1 Zone C — Shared Master, RBAC, Config and Operations

```
-- ☐ ไม่สร้างตาราง stores ใน SBPGI — ใช้ store / mas_store / sevenshop ของระบบ SBP เดิม (API: GET /store/search · /store/list · /store/detail)

CREATE TABLE impacted_stores (
   store_code VARCHAR(5) PRIMARY KEY REFERENCES stores(store_code),
   dv_code VARCHAR(20), opt_dv_user_id VARCHAR(30), latitude NUMERIC(10,7), longitude NUMERIC(10,7),
   is_active BOOLEAN NOT NULL DEFAULT TRUE,
   updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- ☐ ไม่สร้างตาราง workflow_sections ใน SBPGI — ใช้ workflow_state / workflow_route ของ @srm/glb-workflow · วงเงินอนุมัติเก็บใน
common_code (SBPGI_APPROVE_LIMIT)

-- ☐ ไม่สร้างตาราง document_statuses ใน SBPGI — ใช้ workflow_status ของ @srm/glb-workflow

-- ☐ ไม่สร้างตาราง roles ใน SBPGI — ใช้ auth-backend/ABS groups ของระบบ SBP เดิม (ตัดสินใจ 2026-08-05)

-- ☐ ไม่สร้างตาราง menus ใน SBPGI — ใช้ menus/permissions ของ auth-backend (ตัดสินใจ 2026-08-05)

-- ☐ ไม่สร้างตาราง menu_permissions ใน SBPGI — ใช้ permissions ต่อ URL ของ auth-backend (ตัดสินใจ 2026-08-05)

-- ☐ ไม่สร้างตาราง employees ใน SBPGI — ใช้ business_user / business_user_group ของระบบ SBP เดิม

ALTER TABLE impacted_stores
   ADD CONSTRAINT fk_impacted_store_opt_dv FOREIGN KEY (opt_dv_user_id) REFERENCES employees(employee_id);

-- ☐ ไม่สร้างตาราง operator_assignments ใน SBPGI — ใช้ group + scope ของ auth-backend + prepared approvers ของ @srm/glb-workflow
(ตัดสินใจ 2026-08-05)
```

```
CREATE TABLE external_factors (
    factor_code VARCHAR(30) PRIMARY KEY,
    factor_name VARCHAR(200) NOT NULL, factor_remark VARCHAR(500),
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE competitors (
    competitor_code VARCHAR(30) PRIMARY KEY,
    competitor_name VARCHAR(200) NOT NULL,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- ☐ ไม่สร้างตาราง email_templates ใน SBPGI (ตัดสินใจ 2026-08-06) — ใช้ตาราง email_template ของระบบ SBP เดิม (email_template_id · subject_format · body_format) + email_sent

CREATE TABLE status_email_rules (
    status_code VARCHAR(2) NOT NULL REFERENCES document_statuses(status_code),
    template_code VARCHAR(30) NOT NULL REFERENCES email_templates(template_code),
    to_section_code VARCHAR(2) REFERENCES workflow_sections(section_code),
    cc_section_code VARCHAR(2) REFERENCES workflow_sections(section_code),
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    PRIMARY KEY (status_code, template_code)
);

-- ☐ ไม่สร้างตาราง user_accounts ใน SBPGI — ใช้ AWS Cognito + auth-backend — SBPGI รับตัวตนจาก header ของ BFF (ตัดสินใจ 2026-08-05)

CREATE TABLE job_configs (
    job_no VARCHAR(10) PRIMARY KEY,
    job_name VARCHAR(200) NOT NULL,
    cron_expression VARCHAR(100), enabled BOOLEAN NOT NULL DEFAULT TRUE,
    params_json JSONB NOT NULL DEFAULT '{}'::jsonb,
    secret_ref VARCHAR(255), version_no INTEGER NOT NULL DEFAULT 1,
    updated_by VARCHAR(30), updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT ck_job_config_no_inline_secret CHECK (params_json::text !~* '(password|private_key|client_secret)')
);

CREATE TABLE job_run_histories (
    run_id VARCHAR(50) PRIMARY KEY,
    job_no VARCHAR(10) NOT NULL REFERENCES job_configs(job_no),
    period_key VARCHAR(20), status VARCHAR(20) NOT NULL CHECK (status IN
('QUEUED','RUNNING','WAITING','SUCCESS','FAILED','CANCELLED')),
    trigger_type VARCHAR(20) NOT NULL, triggered_by VARCHAR(30), params_snapshot JSONB NOT NULL DEFAULT '{}'::jsonb,
    total_count INTEGER NOT NULL DEFAULT 0, success_count INTEGER NOT NULL DEFAULT 0,
    reject_count INTEGER NOT NULL DEFAULT 0, skipped_count INTEGER NOT NULL DEFAULT 0,
    error_code VARCHAR(80), error_message TEXT,
    started_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP, ended_at TIMESTAMP
);

CREATE TABLE audit_logs (
    id BIGSERIAL PRIMARY KEY,
    table_name VARCHAR(100) NOT NULL, ref_key VARCHAR(200) NOT NULL,
    action_type VARCHAR(80) NOT NULL, old_value JSONB, new_value JSONB,
    reason VARCHAR(500), request_id VARCHAR(80), updated_by VARCHAR(30) NOT NULL,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- ☐ ไม่สร้างตาราง system_configs ใน SBPGI (ตัดสินใจ 2026-08-06) — ใช้ตาราง mas_param ของระบบ SBP เดิม (param_name · param_value · ref_name · description · is_config · active_flag)
```

5.2 Zone A — Impact Pipeline, Sales and Interface

```
CREATE TABLE fgi_impact_processes (
    id BIGSERIAL PRIMARY KEY,
    impacted_store_code VARCHAR(5) NOT NULL REFERENCES impacted_stores(store_code),
    impact_month CHAR(7) NOT NULL,
    process_status VARCHAR(30) NOT NULL, action_status VARCHAR(30),
    last_compensation_amount NUMERIC(14,2),
    workflow_generation_status CHAR(1) NOT NULL DEFAULT 'W' CHECK (workflow_generation_status IN ('W','Y','N')),
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_impact_process UNIQUE (impacted_store_code, impact_month)
);

CREATE TABLE fgi_impact_stores (
    id BIGSERIAL PRIMARY KEY,
    impact_process_id BIGINT NOT NULL REFERENCES fgi_impact_processes(id),
    impacted_store_code VARCHAR(5) NOT NULL REFERENCES impacted_stores(store_code),
    new_store_code VARCHAR(5) NOT NULL REFERENCES stores(store_code),
    impact_month CHAR(7) NOT NULL, distance_km NUMERIC(8,3),
```

```sql
    sales_request_status CHAR(1) NOT NULL DEFAULT 'W' CHECK (sales_request_status IN ('W','P','Y','E')),
    forecast_compensate_percent NUMERIC(7,4), adjust_compensate_percent NUMERIC(7,4),
    forecast_compensation_amount NUMERIC(14,2), adjust_compensation_amount NUMERIC(14,2),
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_impact_store_pair UNIQUE (impacted_store_code, new_store_code, impact_month)
);

CREATE TABLE fgi_impact_sales_summaries (
    id BIGSERIAL PRIMARY KEY,
    impact_process_id BIGINT NOT NULL REFERENCES fgi_impact_processes(id),
    total_working_days INTEGER NOT NULL DEFAULT 0 CHECK (total_working_days >= 0),
    growth_rate_before NUMERIC(9,4), growth_rate_after NUMERIC(9,4), growth_rate_diff NUMERIC(9,4),
    sales_status CHAR(1) NOT NULL DEFAULT 'W' CHECK (sales_status IN ('W','Y','N','E')),
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_sales_summary_process UNIQUE (impact_process_id)
);

CREATE TABLE sales_transactions (
    id BIGSERIAL PRIMARY KEY,
    sales_summary_id BIGINT NOT NULL REFERENCES fgi_impact_sales_summaries(id) ON DELETE CASCADE,
    txn_date DATE NOT NULL, window_no SMALLINT NOT NULL CHECK (window_no BETWEEN 1 AND 4),
    sales_amount NUMERIC(14,2) NOT NULL, sales_diff NUMERIC(14,2),
    is_outlier BOOLEAN NOT NULL DEFAULT FALSE, source_checksum VARCHAR(64) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_sales_day_window UNIQUE (sales_summary_id, txn_date, window_no)
);

CREATE TABLE fgi_impact_competitors (
    id BIGSERIAL PRIMARY KEY,
    impact_process_id BIGINT NOT NULL REFERENCES fgi_impact_processes(id),
    competitor_code VARCHAR(30) NOT NULL REFERENCES competitors(competitor_code),
    name_th VARCHAR(200), branch_th VARCHAR(200), opened_date DATE, closed_date DATE,
    period_key CHAR(7) NOT NULL, updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_impact_competitor UNIQUE (impact_process_id, competitor_code, period_key)
);

CREATE TABLE fcs_qssi_scores (
    id BIGSERIAL PRIMARY KEY,
    store_code VARCHAR(5) NOT NULL REFERENCES stores(store_code),
    category_code VARCHAR(30) NOT NULL, score_period CHAR(7) NOT NULL,
    score_value NUMERIC(10,4) NOT NULL, source_file_name VARCHAR(255), source_checksum VARCHAR(64),
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_qssi_score UNIQUE (store_code, category_code, score_period)
);

CREATE TABLE interface_transactions (
    id BIGSERIAL PRIMARY KEY,
    run_id VARCHAR(50) REFERENCES job_run_histories(run_id),
    data_name VARCHAR(80) NOT NULL, direction VARCHAR(10) NOT NULL CHECK (direction IN ('IN','OUT','INTERNAL')),
    status VARCHAR(20) NOT NULL CHECK (status IN ('READY','SENT','ACKED','COMPLETED','FAILED','FAILED_RETRY')),
    impact_process_id BIGINT REFERENCES fgi_impact_processes(id),
    sales_summary_id BIGINT REFERENCES fgi_impact_sales_summaries(id),
    doc_no VARCHAR(10), business_key VARCHAR(200) NOT NULL, period_key VARCHAR(20) NOT NULL,
    correlation_id VARCHAR(100), file_name VARCHAR(255), file_checksum VARCHAR(64),
    outbox_status VARCHAR(20), return_code VARCHAR(50), return_message VARCHAR(500),
    retry_count INTEGER NOT NULL DEFAULT 0, sent_at TIMESTAMP, acked_at TIMESTAMP,
    purge_after TIMESTAMP, legal_hold BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP, completed_at TIMESTAMP,
    CONSTRAINT uq_interface_business UNIQUE (data_name, direction, business_key, period_key),
    CONSTRAINT ck_interface_typed_reference CHECK (num_nonnulls(impact_process_id, sales_summary_id, doc_no) >= 1)
);
```

5.3 Zone B — Document and Internal Workflow

```sql
CREATE TABLE compensation_documents (
    doc_no VARCHAR(10) PRIMARY KEY,
    year INTEGER NOT NULL, running_no INTEGER NOT NULL,
    impact_process_id BIGINT NOT NULL UNIQUE REFERENCES fgi_impact_processes(id),
    impacted_store_code VARCHAR(5) NOT NULL REFERENCES impacted_stores(store_code),
    impact_month CHAR(7), new_store_code VARCHAR(5) REFERENCES stores(store_code), round_no INTEGER,
    source VARCHAR(20) NOT NULL DEFAULT 'FS' CHECK (source IN ('FS','MANUAL')),
    status_code VARCHAR(2) NOT NULL REFERENCES document_statuses(status_code),
    current_section_code VARCHAR(2) REFERENCES workflow_sections(section_code),
    total_compensation_amount NUMERIC(14,2) NOT NULL DEFAULT 0,
    version_no INTEGER NOT NULL DEFAULT 1,
    created_by VARCHAR(30) NOT NULL, created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(30), updated_at TIMESTAMP,
    CONSTRAINT uq_comp_year_running UNIQUE (year, running_no),
    CONSTRAINT uq_comp_business UNIQUE (source, impacted_store_code, impact_month, new_store_code, round_no)
);

ALTER TABLE interface_transactions
```

```sql
  ADD CONSTRAINT fk_interface_doc_no FOREIGN KEY (doc_no) REFERENCES compensation_documents(doc_no);

CREATE TABLE document_new_stores (
  id BIGSERIAL PRIMARY KEY,
  doc_no VARCHAR(10) NOT NULL REFERENCES compensation_documents(doc_no) ON DELETE CASCADE,
  new_store_code VARCHAR(5) NOT NULL REFERENCES stores(store_code),
  distance_km NUMERIC(8,3), compensate_percent NUMERIC(7,4) NOT NULL CHECK (compensate_percent BETWEEN 0 AND 100),
  compensation_amount NUMERIC(14,2) NOT NULL DEFAULT 0,
  source_system VARCHAR(30) NOT NULL, updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT uq_doc_new_store UNIQUE (doc_no, new_store_code)
);

CREATE TABLE document_competitors (
  id BIGSERIAL PRIMARY KEY,
  doc_no VARCHAR(10) NOT NULL REFERENCES compensation_documents(doc_no) ON DELETE CASCADE,
  competitor_code VARCHAR(30) NOT NULL REFERENCES competitors(competitor_code),
  name_th VARCHAR(200), branch_th VARCHAR(200), opened_date DATE, closed_date DATE, impact_date DATE,
  detail TEXT, remark TEXT, source_system VARCHAR(30) NOT NULL,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT uq_doc_competitor UNIQUE (doc_no, competitor_code)
);

CREATE TABLE document_external_factors (
  id BIGSERIAL PRIMARY KEY,
  doc_no VARCHAR(10) NOT NULL REFERENCES compensation_documents(doc_no) ON DELETE CASCADE,
  factor_code VARCHAR(30) NOT NULL REFERENCES external_factors(factor_code),
  date_from DATE, date_to DATE, detail TEXT, remark TEXT,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT uq_doc_factor UNIQUE (doc_no, factor_code, date_from),
  CONSTRAINT ck_doc_factor_dates CHECK (date_to IS NULL OR date_from IS NULL OR date_to >= date_from)
);

CREATE TABLE consideration_logs (
  id BIGSERIAL PRIMARY KEY,
  doc_no VARCHAR(10) NOT NULL REFERENCES compensation_documents(doc_no),
  section_code VARCHAR(2) NOT NULL REFERENCES workflow_sections(section_code),
  result VARCHAR(100) NOT NULL, result_category VARCHAR(50), detail TEXT,
  consider_by VARCHAR(30) NOT NULL REFERENCES employees(employee_id),
  action_datetime TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP, request_id VARCHAR(80)
);

CREATE TABLE document_attachments (
  attach_id BIGSERIAL PRIMARY KEY,
  doc_no VARCHAR(10) NOT NULL REFERENCES compensation_documents(doc_no),
  section_code VARCHAR(2) NOT NULL REFERENCES workflow_sections(section_code),
  file_name VARCHAR(255) NOT NULL, mime_type VARCHAR(100) NOT NULL,
  file_size BIGINT NOT NULL CHECK (file_size <= 5242880),
  storage_provider VARCHAR(30) NOT NULL, bucket VARCHAR(120) NOT NULL,
  object_key VARCHAR(500) NOT NULL, sha256 VARCHAR(64) NOT NULL,
  scan_status VARCHAR(20) NOT NULL CHECK (scan_status IN ('PENDING','CLEAN','BLOCKED','FAILED')),
  scanned_at TIMESTAMP, scan_message VARCHAR(500), uploaded_by VARCHAR(30) NOT NULL,
  uploaded_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP, deleted_flag CHAR(1) NOT NULL DEFAULT 'N',
  CONSTRAINT uq_doc_attachment_hash UNIQUE (doc_no, sha256, deleted_flag)
);

CREATE TABLE compensation_histories (
  id BIGSERIAL PRIMARY KEY,
  store_code VARCHAR(5) NOT NULL REFERENCES stores(store_code),
  ref_doc_no VARCHAR(10) REFERENCES compensation_documents(doc_no),
  submit_account_month CHAR(7) NOT NULL, compensate_amount NUMERIC(14,2) NOT NULL,
  accounting_status VARCHAR(30), external_ref VARCHAR(100),
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT uq_compensation_history UNIQUE (store_code, ref_doc_no, submit_account_month)
);

-- ☐ ไม่สร้างตาราง workflow_instances ใน SBPGI (ตัดสินใจ 2026-08-06) — ใช้ workflow_transaction ของ @srm/glb-workflow

-- ☐ ไม่สร้างตาราง workflow_tasks ใน SBPGI (ตัดสินใจ 2026-08-06) — ใช้ workflow_approver/workflow_transaction ของ @srm/glb-workflow
```

5.4 Required Indexes, Partial Uniqueness and Purge

```sql
CREATE INDEX idx_document_status_section ON compensation_documents(status_code, current_section_code);
CREATE INDEX idx_document_impact_process ON compensation_documents(impact_process_id);
CREATE INDEX idx_task_doc_status ON workflow_tasks(doc_no, task_status);
CREATE UNIQUE INDEX uq_task_one_open_section ON workflow_tasks(instance_id, section_code) WHERE task_status = 'OPEN';
CREATE INDEX idx_consideration_timeline ON consideration_logs(doc_no, action_datetime DESC);
CREATE INDEX idx_interface_pending ON interface_transactions(data_name, status, sent_at);
CREATE INDEX idx_interface_impact_process ON interface_transactions(impact_process_id);
CREATE INDEX idx_interface_sales_summary ON interface_transactions(sales_summary_id);
CREATE INDEX idx_interface_doc ON interface_transactions(doc_no);
CREATE UNIQUE INDEX uq_job_running ON job_run_histories(job_no, COALESCE(period_key, '')) WHERE status = 'RUNNING';
CREATE INDEX idx_audit_ref ON audit_logs(table_name, ref_key, updated_at DESC);
```

```
-- Retention worker: delete only terminal, expired, non-held rows in bounded batches.
WITH purge_candidates AS (
  SELECT id FROM interface_transactions
  WHERE status IN ('ACKED', 'COMPLETED')
    AND purge_after < CURRENT_TIMESTAMP
    AND legal_hold = FALSE
    AND data_name = ANY(:data_names)
  ORDER BY id
  LIMIT :batch_size
  FOR UPDATE SKIP LOCKED
)
DELETE FROM interface_transactions i
USING purge_candidates p
WHERE i.id = p.id
RETURNING i.id, i.data_name, i.business_key;
```

6. Index and Constraint Plan

Table	Index / constraint	Reason
compensation_documents	UNIQUE (year, running_no), UNIQUE(source, impacted_store_code, impact_month, new_store_code, round_no), INDEX(status_code,current_section_code), INDEX(impact_process_id)	docNo uniqueness, duplicate guard, list/inbox/report, pipeline trace
workflow_tasks	INDEX(doc_no, task_status), INDEX(section_code, task_status), UNIQUE(instance_id, section_code, task_status) filtered OPEN	current task guard and inbox
document_new_stores	INDEX(doc_no), CHECK compensate_percent between 0 and 100	detail load and allocation validation
consideration_logs	INDEX(doc_no, action_datetime DESC), INDEX(result_category)	timeline/report result filter
document_attachments	INDEX(doc_no), INDEX(scan_status), UNIQUE(sha256, doc_no, deleted_flag)	attachment list/download/security
job_run_histories	INDEX(job_no, period, status), UNIQUE(job_no, period) filtered RUNNING	manual run concurrency guard
audit_logs	INDEX(table_name, ref_key), INDEX(updated_at DESC)	admin history search
interface_transactions	INDEX(data_name,status), INDEX(impact_process_id), INDEX(doc_no)	tracking and pending ACK

7. Transaction Rules

Use case	Transaction boundary	Rollback rule
Create document	docNo sequence lock + compensation_documents + workflow instance/task + audit	any fail rollback all; no partial document
Submit action	lock current OPEN task + insert consideration_logs + close/open task + update document	duplicate/current task conflict returns 409
Attachment upload	metadata insert only after storage write and AV clean; objectKey never exposed	storage/scan fail leaves no CLEAN metadata
Job 4 IAS request	durable file (fsync + atomic rename + checksum) ก่อน transaction W→P + outbox READY	file fail คง W; DB fail rollback W→P/outbox; SFTP fail retry transaction เดิม
Interface ACK/purge	ACK compare-and-set บน transaction เดิม; purge เฉพาะ terminal + purge_after + non-held	pending/failed/unacked/legal-hold ห้ามลบ

Use case	Transaction boundary	Rollback rule
Job manual run	acquire run lock + job_run_histories RUNNING before processing	fatal fail marks run FAILED and keeps record-level rejects
Master/config/email mutation	validate reason + update entity + insert audit_logs	audit fail rollback mutation

8. Seed Data

Domain	Required seed
workflow_sections	06, 08, 01, 02, 03 only
document_statuses	6 statuses: 5 waiting statuses + completed
roles	00, 01, 02, 03, 04, 05, 06, 10 per RBAC model
email_templates	EM-01..EM-08
system_configs	impact radius 1/2 km, workflow.avp_amount_threshold=100000, sales data threshold=60, growth rate threshold=-10
job_configs	Jobs 1-10 and 8b with enabled/schedule/default params as schema reference

9. Migration and Verification Checklist

Area	Check
Naming	all new tables/columns lower_snake_case
Leading zero	store_code/new_store_code stored as VARCHAR(5), never numeric
docNo	year/running_no/doc_no generated in DB transaction; concurrency test 20 parallel requests
Workflow	no active 04/05 accounting sections/statuses
Security	no secrets in system_configs/job_configs; storage objectKey not returned to FE
External interface	credential/certificate/private key อยู่ Secret Manager ผ่าน secretRef; TLS verify-full หรือ SFTP strict known_hosts; ทดสอบ rotation และ invalid certificate/host key
Tracking retention	backfill typed FK/purge_after, validate FK, dry-run count แล้ว purge เฉพาะ ACKED/COMPLETED เป็น batch; reconcile count ก่อน/หลัง
Data integrity	FK/check constraints enabled before SIT; reject legacy invalid enum values
Performance	list/report/inbox queries explain plan uses indexes above

10. Related LLDD

Document	DB usage
LLDD-BE-API-Dashboard-Summary.pdf	workflow_transaction / workflow_approver (@srm/glb-workflow)(R), compensation_documents(R), compensation_histories(R), fgi_impact_sales_summaries(R)
LLDD-BE-API-Document-List-Search.pdf	workflow_transaction / workflow_approver (@srm/glb-workflow)(R), compensation_documents(R), impacted_stores(R), fgi_impact_sales_summaries(R)
LLDD-BE-API-Document-Create-Update.pdf	compensation_documents(R/W), workflow_transaction / workflow_approver (@srm/glb-workflow)(W), document_new_stores(R/W), document_competitors(R/W)

Document	DB usage
LLDD-BE-API-Document-Detail-Aggregate.pdf	compensation_documents(R), impacted_stores(R), document_new_stores(R), document_competitors(R)
LLDD-BE-API-Document-Workflow-Actions.pdf	workflow_transaction / workflow_history / workflow_approver (@srm/glb-workflow)(R/W), compensation_documents(W), consideration_logs(W), status_email_rules(R)
LLDD-BE-API-Workflow-Instances.pdf	fgi_impact_processes / fgi_impact_stores(R/W), compensation_documents(R/W), workflow_transaction (@srm/glb-workflow)(R/W), workflow_approver (@srm/glb-workflow)(W)
LLDD-BE-API-Attachment-Sales-Timeline.pdf	document_attachments(R/W), compensation_documents(R), fgi_impact_sales_summaries(R), sales_transactions(R)
LLDD-BE-API-Lookup-RBAC-Email.pdf	stores / impacted_stores(R), document_statuses / workflow_sections(R), employees(R), roles / menus / menu_permissions(R/W)
LLDD-BE-API-Report-Master-Config.pdf	compensation_documents(R), compensation_histories(R), consideration_logs(R), operator_assignments(R/W)
LLDD-BE-Job-Batch-Email-SRM.pdf	job_configs(R/W), job_run_histories(R/W), interface_transactions(R/W), email_template (SBP)(R/W)
LLDD-BE-Job-1-ImportQSSI.pdf	fcs_qssi_scores(W)
LLDD-BE-Job-2-ImportImpactStore.pdf	fgi_impact_stores(W)
LLDD-BE-Job-3-ImportImpactCompetitor.pdf	fgi_impact_competitors(W)
LLDD-BE-Job-4-PrepareImpactStoreToIAS.pdf	fgi_impact_stores(R/W), fgi_impact_sales_summaries(R/W), interface_transactions(W), job_run_histories(W)
LLDD-BE-Job-5-ImportImpactSaleFromIAS.pdf	sales_transactions(W), fgi_impact_sales_summaries(R/W), interface_transactions(W)
LLDD-BE-Job-6-ExportImpactStoreToFS.pdf	fgi_impact_processes(R/W), fgi_impact_stores(R/W), fcs_qssi_scores(R), interface_transactions(W)
LLDD-BE-Job-7-SyncCompetitorToDocument.pdf	fgi_impact_competitors(R), compensation_documents(R), document_competitors(W), interface_transactions(W)
LLDD-BE-Job-8-CreateCompensationDocument.pdf	fgi_impact_stores(R/W), fgi_impact_processes(R), compensation_documents(W), interface_transactions(W)
LLDD-BE-Job-8b-StartInternalWorkflow.pdf	fgi_impact_stores(R/W), compensation_documents(R/W), workflow_instances(W), workflow_tasks(W)
LLDD-BE-Job-9-SyncNewStoreToDocument.pdf	fgi_impact_stores(R), compensation_documents(R), document_new_stores(W), interface_transactions(W)
LLDD-BE-Job-10-NotifyNoReceiveData.pdf	interface_transactions(R), email_templates(R), status_email_rules(R)